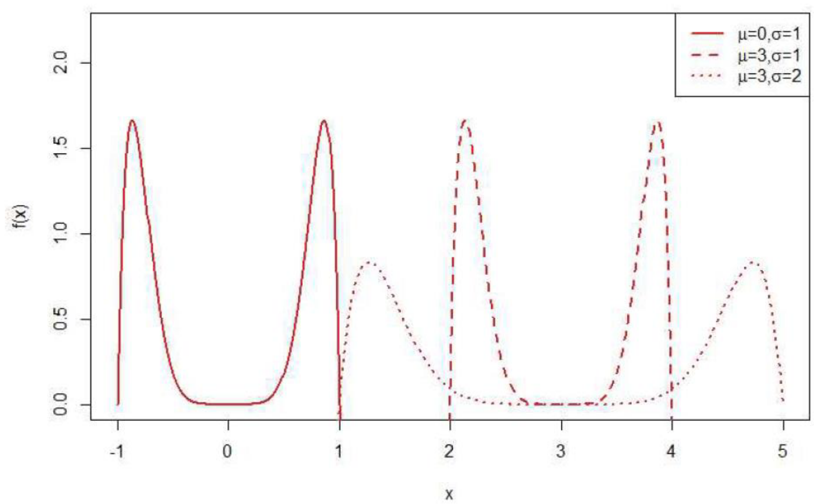
CB 3.36 Consider the pdf $f(x) = \frac{63}{4}(x^6 - x^8)$, $-1 < x < 1$. Graph $(1/\sigma)f((x-\mu)/\sigma)$ for each of the following on the same axes.

(a) $\mu = 0$, $\sigma = 1$

(b) $\mu = 3$, $\sigma = 1$

(c) $\mu = 3$, $\sigma = 2$




```
f <- function(x,mu,sigma) {
(1/sigma)*(63/4)*(((x-mu)/sigma)^6-((x-mu)/sigma)^8)
}

x1 <- seq(-1,1,length=200)
plot(x1,f(x1,0,1),type='l',xlim=c(-1,5))

x2 <- x1*1+3
lines(x2,f(x2,3,1),col='red')

x3 <- x1*2+3
lines(x3,f(x3,3,2),col='blue')
```

3.39

$$f(x|\mu,\sigma) = \frac{1}{\sigma\pi\left(1+\left(\frac{x-\mu}{\sigma}\right)^2\right)}$$

$z = \frac{x-\mu}{\sigma}$

$x = z\sigma + \mu$

(a.) $f(z|\mu=0,\sigma=1) = \frac{1}{\pi(1+x^2)}$

$$\int_{-\infty}^{m} \frac{1}{\pi(1+x^2)}\,dx = .5$$

$$\Rightarrow \frac{1}{\pi}\arctan(x)\Big|_{-\infty}^{m} = .5$$

$\arctan(m) - \arctan(-\infty) = .5\pi$

$\arctan(m) + \frac{\pi}{2} = \frac{\pi}{2}$

$\arctan(m) = 0$

$m = \tan(0)$

$\boxed{m = 0 \quad \text{if} \quad \mu = 0 ; \sigma = 1}$

$x_m = z_m\sigma + \mu$

$x_m = 0 + \mu$

$\boxed{x_m = \mu \quad \text{if} \quad \mu = \mu ; \sigma = \sigma}$

3.39(b.) $P(x \le \mu - \sigma) = \frac{1}{4}$ and $P(x \ge \mu + \sigma) = \frac{1}{4}$

First let $\mu = 0$, $\sigma = 1$

show

$P(x \le -1) = \frac{1}{4}$

$$\int_{-\infty}^{-1} \frac{1}{\pi(1+x^2)}\,dx = \frac{1}{\pi}\arctan(x)\Big|_{-\infty}^{-1} = \frac{1}{\pi}\left[\arctan(-1) - \arctan(-\infty)\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{4} + \frac{\pi}{2}\right] = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

show $P(x \ge 1) = \frac{1}{4}$

$$\int_{1}^{\infty} \frac{1}{\pi}\frac{1}{(1+x^2)}\,dx = \frac{1}{\pi}\arctan(x)\Big|_{1}^{\infty} = \frac{1}{\pi}\left[\arctan(\infty) - \arctan(1)\right]$$

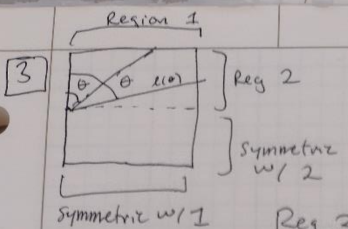
$$= \frac{1}{\pi}\left[\frac{\pi}{2} - \frac{\pi}{4}\right] = \frac{1}{4}$$

If $\mu = \mu$ and $\sigma = \sigma$ then $x_\alpha = \sigma z_\alpha + \mu$

Q1: $z_{Q1} = -1$ $\quad x_{Q1} = \sigma(-1) + \mu = \mu - \sigma$

Q3: $z_{Q3} = 1$ $\quad x_{Q3} = \sigma(1) + \mu = \mu + \sigma$

3



Reg 1

$$\cos(\theta) = \frac{1/2}{\ell(\theta)}$$

$$\boxed{\ell(\theta) = \frac{1}{2\cos(\theta)}}$$

Reg 2

$$\cos\left(\frac{\pi}{2}-\theta\right) = \sin(\theta) = \frac{1}{\ell_n(\theta)} \Rightarrow \boxed{\ell(\theta) = \frac{1}{\sin(\theta)}}$$

$$E[\ell(\theta)] = 2\int_0^{\tan^{-1}(2)} \frac{1}{\pi}\frac{1}{2\cos(\theta)}\,d\theta + 2\int_{\tan^{-1}(2)}^{\pi/2} \frac{1}{\pi}\frac{1}{\sin(\theta)}\,d\theta \approx 0.76587$$

$$= \frac{2}{\pi}\left[\frac{1}{2}\log\left|\tan(\theta)+\sec(\theta)\right|\Big|_0^{\tan^{-1}(2)} + \log\left|\cot(\theta) - \csc(\theta)\right|\Big|_{\tan^{-1}(2)}^{\frac{\pi}{2}}\right]$$

This is fine

$$= \frac{2}{\pi}\left[\frac{1}{2}\ln\left(2+\sec(\tan^{-1}(2))\right) - \ln(1)\right] + \frac{2}{\pi}\left[\log\left|\csc\left(\frac{\pi}{2}\right) - \cot\left(\frac{\pi}{2}\right)\right| - \log\left|\csc(\tan^{-1}(2)) - \cot(\tan^{-1}(2))\right|\right]$$

This is f

$$= \frac{1}{\pi}\ln\left(2+\sec(\tan^{-1}(2))\right) + \frac{2}{\pi}\left[\log\left|\csc\left(\frac{\pi}{2}\right) - \cot\left(\frac{\pi}{2}\right)\right| - \log\left|\csc\tan^{-1}(2) - \cot\tan^{-1}(2)\right|\right]$$

$$= \frac{1}{\pi}\ln\left[2+\sqrt{5}\right] + \frac{2}{\pi}\left[\log(1+0) - \log\left|\frac{\sqrt{5}}{2} - \frac{1}{2}\right|\right]$$

$$= \frac{1}{\pi}\left[\ln\left[2+\sqrt{5}\right] - 2\log\left|\frac{\sqrt{5}}{2} - \frac{1}{2}\right|\right]$$

$$= \frac{1}{\pi}\left[\ln\left[2+\sqrt{5}\right] + 2\log\left(\frac{2}{\sqrt{5}-1}\right)\right]$$

Any of these are fine.

(b)

	estimate	standard_error
1000	0.7619435	0.0070537585
10000	0.7665389	0.0022399718
100000	0.7655717	0.0007075016

5. 4. (a) – (d) see code. The right answer is log(26)/(10 * atan(5)), which is about 0.23723. The answers in (a) – (d) just need to be close. The numbers I give here won't be the exactly the same as their answers, but they'll all be pretty close.

A. 0.2362374
B. 0.2406166
C. 0.2334262
D. 0.2338831

E. I don't expect them to calculate standard errors. The answer should be either A or D, but I think you can give full credit for any thoughtful answer.

The issue with (B) is that you throw away a lot of draws if you just use a Cauchy. If they use a truncated Cauchy from an R package, then B is probably best. The importance distribution in (C) is a poor match for f(). (A) doesn't match f() very well either. (D) is a better match for f() in most places, but its tail (near 1) is too weak. If they compute this, they should find A.

CB 4.1 A random point (X, Y) is distributed uniformly on the square with vertices $(1, 1)$, $(1, -1)$, $(-1, 1)$, and $(-1, -1)$. That is, the joint pdf is $f(x, y) = \frac{1}{4}$ on the square. Determine the probabilities of the following events.

(a) $X^2 + Y^2 < 1$
The circle $x^2 + y^2 \leq 1$ has area π, so $P(X^2 + Y^2 \leq 1) = \frac{\pi}{4}$

(b) $2X - Y > 0$
The area below the line $y = 2x$ is half of the area of the square, so $P(2X - Y > 0) = \frac{1}{2}$

(c) $|X + Y| < 2$
Clearly $P(|X + Y| < 2) = 1$.

CB 4.4 A pdf is defined by

$$f(x, y) = \begin{cases} C(x + 2y) & \text{if } 0 < y < 1 \text{ and } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the value of C.

$$\int_0^1 \int_0^2 C(x + 2y)dxdy = 4C = 1$$

Thus $C = \frac{1}{4}$

(b) Find the marginal distribution of X.

$$f_X(x) = \begin{cases} \int_0^1 \frac{1}{4}(x + 2y)dy = \frac{1}{4}(x + 1) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

(c) Find the joint cdf of X and Y.
$F_{XY}(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(v, u)dvdu$. The way this integral is calculated depends on the values of x and y. For example, for $0 < x < 2$ and $0 < y < 1$,

$$F_{XY}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)dvdu = \int_0^x \int_0^y \frac{1}{4}(u + 2v)dvdu = \frac{x^2 y}{8} + \frac{y^2 x}{4}$$

But for $0 < x < 2$ and $1 \leq y$,

$$F_{XY}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)dvdu = \int_0^x \int_0^1 \frac{1}{4}(u + 2v)dvdu = \frac{x^2}{8} + \frac{x}{4}$$

The complete definition of F_{XY} is

$$F_{XY} = \begin{cases} 0 & x \leq 0 \text{ or } y \leq 0 \\ x^2 y/8 + y^2 x/4 & 0 < x < 2 \text{ and } 0 < y < 1 \\ y/2 + y^2/2 & 2 \leq x \text{ and } 0 < y < 1 \\ x^2/8 + x/4 & 0 < x < 2 \text{ and } 1 \leq y \\ 1 & 2 \leq x \text{ and } 1 \leq y \end{cases}$$

(d) Find the pdf of the random variable $Z = 9/(X + 1)^2$.
The function $z = g(x) = 9/(x + 1)^2$ is monotone on $0 < x < 2$, so use Theorem 2.1.5 to obtain $f_Z(z) = 9/(8z^2)$, $1 < z < 9$.

CB 4.5

(a) Find $P(X > \sqrt{Y})$ if X and Y are jointly distributed with pdf

$$f(x,y) = x + y \qquad 0 \le x \le 1; 0 \le y \le 1.$$

$$P(X > \sqrt{Y}) = \int_0^1 \int_{\sqrt{y}}^1 (x+y)dxdy = \frac{7}{20}$$

(b) Find $P(X^2 < Y < X)$ if X and Y are jointly distributed with pdf

$$f(x,y) = 2x, \qquad 0 \le x \le 1; 0 \le y \le 1.$$

$$P(X^2 < Y < X) = \int_0^1 \int_y^{\sqrt{y}} 2xdxdy = \frac{1}{6}$$

CB 4.7 A woman leaves for work between 8 AM and 8:30 AM and takes between 40 and 50 minutes to get there. Let the random variable X denote her time of departure, and the random variable Y the travel time. Assuming that these variables are independent and uniformly distributed, find the probability that the woman arrives at work before 9 AM.

We will measure time in minutes past 8 A.M. So $X \sim uniform(0,30), Y \sim unifrom(40,50)$ and the joint pdf is $1/300$ on the rectangle $(0,30) \times (40,50)$.

$$P(\text{arrive before 9 A.M.}) = P(X + Y < 60) = \int_{40}^{50} \int_0^{60-y} \frac{1}{300} dxdy = \frac{1}{2}$$

9. (a)

(i) $P(X > Y) = \int_0^1 \int_0^x \frac{6}{7}(x+y)^2 dydx = \int_0^1 [\frac{6}{21}(x+y)^3 |_0^x] dx = \int_0^1 \frac{6}{21}(2x)^3 - \frac{6}{21}x^3 dx = \int_0^1 \frac{16}{7}x^3 - \frac{2}{7}x^3 dx$

$= \int_0^1 \frac{14}{7} x^3 dx = \frac{14}{28} x^4 |_0^1 = \boxed{\frac{1}{2}}$

(ii) $P(X + Y \le 1) = P(X \le 1 - Y) = \int_0^1 \int_0^{1-y} \frac{6}{7}(x+y)^2 dxdy = \int_0^1 [\frac{6}{21}(x+y)^3 |_0^{1-y}] dy = \int_0^1 \frac{6}{21} - \frac{6}{21} y^3 dy$

$= \frac{6}{21} y - \frac{6}{84} y^4 |_0^1 = \frac{6}{21} - \frac{6}{84} = \boxed{\frac{3}{14}}$

(iii) $P(X \le \frac{1}{2}) = \int_0^{1/2} \int_0^1 \frac{6}{7}(x+y)^2 dydx = \int_0^{1/2} [\frac{2}{7}(x+y)^3 |_0^1] = \int_0^{1/2} \frac{2}{7}(x+1)^3 - \frac{2}{7}x^3 dx$

$= \frac{2}{28}(x+1)^4 - \frac{2}{28}x^4 |_0^{1/2}$

$= \frac{1}{14}(\frac{3}{2})^4 - \frac{1}{14}(\frac{1}{2})^4 - \frac{1}{14} = \boxed{\frac{2}{7}}$

(b) $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_0^1 \frac{6}{7}(x+y)^2 dy = \frac{2}{7}(x+y)^3 |_0^1 = \frac{2}{7}(x+1)^3 - \frac{2}{7}x^3 = \frac{2}{7}(3x^2 + 3x + 1)$

$= \boxed{\frac{2}{7}((x+1)^3 - x^3) \text{ or } \frac{2}{7}(3x^2 - 3x + 1) \quad 0 \le x \le 1}$

similarly, $\boxed{f_Y(y) = \frac{2}{7}((y+1)^3 - y^3) \quad 0 \le y \le 1}$

10.

4. CB 4.31 Suppose that the random variable Y has a binomial distribution with n trials and success probability X, where n is a given constant and X is a uniform(0,1) random variable.

(a) Find EY and Var Y.

$$\mathrm{E}Y = \mathrm{E}\{\mathrm{E}(Y|X)\} = \mathrm{E}nX = \frac{n}{2}.$$

$$\mathrm{Var}Y = \mathrm{Var}(\mathrm{E}(Y|X)) + \mathrm{E}(\mathrm{Var}(Y|X)) = \mathrm{Var}(nX) + \mathrm{E}nX(1-X) = \frac{n^2}{12} + \frac{n}{6}.$$

(b) Find the joint distribution of X and Y.

$$P(Y = y, X \leq x) = \binom{n}{y} x^y (1-x)^{n-y}, \quad y = 0, 1, \ldots, n, \quad 0 < x < 1.$$

(c) Find the marginal distribution of Y

$$P(Y = y) = \binom{n}{y} \frac{\Gamma(y+1)\Gamma(n-y+1)}{\Gamma(n+2)}.$$

11.

(a.) $X|Y \sim Bin(n,Y) \quad Y \sim Beta(a,b)$

$$f(x) = \int_0^1 f_{XY}(x,y)\,dy = \int_0^1 f(x|y)f(y)\,dy$$

$$= \int_0^1 \binom{n}{x} y^x (1-y)^{n-x} \frac{\Gamma(a,b)}{\Gamma(a)\Gamma(b)} y^{a-1}(1-y)^{b-1}\,dy$$

$$= \binom{n}{x}\frac{\Gamma(a,b)}{\Gamma(a)\Gamma(b)} \int_0^1 y^{a+x-1}(1-y)^{b+n-x-1}\,dy$$

$$= \binom{n}{x}\frac{\Gamma(a,b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(a+x)\Gamma(b+n-x)}{\Gamma(a+b+n)} \quad x=0,1,\dots,n$$

This is known as the Beta-Binomial distribution

(b.) $X|Y \sim Exp(Y) \quad Y \sim Gamma(a,b)$

$$f_X(x) = \int_0^\infty y e^{-xy} \frac{b^a}{\Gamma(a)} y^{a-1} e^{-yb}\,dy$$

$$= \frac{b^a}{\Gamma(a)} \int_0^\infty y^{a+1-1} e^{-y(x+b)}\,dy$$

$$= \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+1)}{(x+b)^{a+1}} = \boxed{\frac{ab^a}{(x+b)^{a+1}},\ x>0}$$

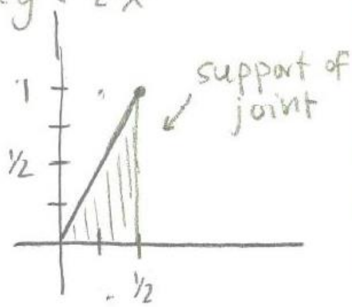
12.

$f_X(x) = 24x^2 \quad 0<x<\frac{1}{2} \qquad f(y|x) = \frac{y}{2x^2},\ 0<y<2x$

$$F_{XY}(x,y) = (24x^2)\left(\frac{y}{2x^2}\right) = 12y \quad 0<x<\tfrac{1}{2},\ 0<y<2x<1$$

$$f_Y(y) = \int_{y/2}^{1/2} 12y\,dx = 12yx\Big|_{y/2}^{1/2} = 12y\left(\tfrac{1}{2}\right) - 12y\left(\tfrac{y}{2}\right)$$

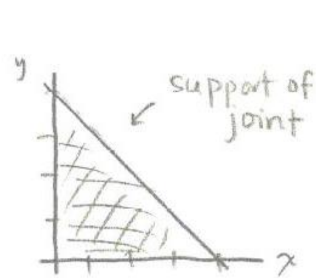
$$= 6y - 6y^2,\ 0<y<1$$



$$f(x|y) = \frac{12y}{6y(1-y)} = \boxed{\frac{2}{1-y},\ \frac{y}{2}<x<\frac{1}{2}}$$

13.

$$f_{XY}(x,y) = 24xy \qquad x>0,\ y>0,\ 0<x+y<1$$



$$f_Y(y) = \int_0^{1-y} 24xy\,dx = 24y \cdot \frac{1}{2}x^2 \Big|_0^{1-y}$$

$$= 12y(1-y)^2 \quad 0<y<1$$

$$f(x|y) = \frac{24xy}{12y(1-y)^2} = \frac{2x}{(1-y)^2}$$

$$P\left(X<\frac{1}{2} \,\middle|\, Y=\frac{1}{4}\right) = \int_0^{1/2} \frac{2x}{(1-\frac{1}{4})^2}\,dx = \frac{16}{9}\cdot x^2\Big|_0^{1/2}$$

$$= \left(\frac{16}{9}\right)\left(\frac{1}{2}\right)^2 = \boxed{\frac{4}{9}}$$

14.

	$x>1$	$y>0$

$$f(x|Y) = y x^{-(y+1)} \qquad f_Y(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}$$

$$f_X(x) = \int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} y x^{-(y+1)} y^{\alpha-1} e^{-\beta y}\,dy = \frac{\beta^\alpha}{\Gamma(\alpha)} \int y^{(\alpha-1)+1} e^{\log(x)^{-(y+1)}} e^{-\beta y} dy$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty y^{(\alpha+1)-1} e^{-(y+1)\log x - \beta y}\,dy$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty y^{(\alpha+1)-1} e^{-\log x} e^{-y(\log x+\beta)}\,dy$$

$$= \frac{\beta^\alpha e^{\log x^{-1}}}{\Gamma(\alpha)} \int_0^\infty \underbrace{y^{(\alpha+1)-1} e^{-y(\log x+\beta)}}_{\text{kernel of gamma }(\alpha+1,\ \log x+\beta)}\,dy$$

$$= \frac{\beta^\alpha}{x\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+1)}{(\log x+\beta)^{\alpha+1}} =$$

$$f_X(x) = \frac{\alpha\beta^\alpha}{x(\log x+\beta)^{\alpha+1}} \quad \text{for } x>1$$

3.24 a. $f_x(x) = \frac{1}{\beta}e^{-x/\beta}$, $x > 0$. For $Y = X^{1/\gamma}$, $f_Y(y) = \frac{\gamma}{\beta}e^{-y^\gamma/\beta}y^{\gamma-1}$, $y > 0$. Using the transformation $z = y^\gamma/\beta$, we calculate

$$\mathrm{E}Y^n = \frac{\gamma}{\beta}\int_0^\infty y^{\gamma+n-1}e^{-y^\gamma/\beta}dy = \beta^{n/\gamma}\int_0^\infty z^{n/\gamma}e^{-z}dz = \beta^{n/\gamma}\Gamma\left(\frac{n}{\gamma}+1\right).$$

Thus $\mathrm{E}Y = \beta^{1/\gamma}\Gamma(\frac{1}{\gamma}+1)$ and $\mathrm{Var}Y = \beta^{2/\gamma}\left[\Gamma\left(\frac{2}{\gamma}+1\right) - \Gamma^2\left(\frac{1}{\gamma}+1\right)\right]$.

b. $f_x(x) = \frac{1}{\beta}e^{-x/\beta}$, $x > 0$. For $Y = (2X/\beta)^{1/2}$, $f_Y(y) = ye^{-y^2/2}$, $y > 0$. We now notice that

$$\mathrm{E}Y = \int_0^\infty y^2 e^{-y^2/2}dy = \frac{\sqrt{2\pi}}{2}$$

since $\frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty y^2 e^{-y^2/2} = 1$, the variance of a standard normal, and the integrand is symmetric. Use integration-by-parts to calculate the second moment

$$\mathrm{E}Y^2 = \int_0^\infty y^3 e^{-y^2/2}dy = 2\int_0^\infty ye^{-y^2/2}dy = 2,$$

where we take $u = y^2$, $dv = ye^{-y^2/2}$. Thus $\mathrm{Var}Y = 2(1 - \pi/4)$.

c. The gamma(a, b) density is

$$f_X(x) = \frac{1}{\Gamma(a)b^a}x^{a-1}e^{-x/b}.$$

Make the transformation $y = 1/x$ with $dx = -dy/y^2$ to get

$$f_Y(y) = f_X(1/y)|1/y^2| = \frac{1}{\Gamma(a)b^a}\left(\frac{1}{y}\right)^{a+1}e^{-1/by}.$$

The first two moments are

$$\begin{aligned}
\mathrm{E}Y &= \frac{1}{\Gamma(a)b^a}\int_0^\infty \left(\frac{1}{y}\right)^a e^{-1/by} = \frac{\Gamma(a-1)b^{a-1}}{\Gamma(a)b^a} = \frac{1}{(a-1)b} \\
\mathrm{E}Y^2 &= \frac{\Gamma(a-2)b^{a-2}}{\Gamma(a)b^a} = \frac{1}{(a-1)(a-2)b^2},
\end{aligned}$$

and so $\mathrm{Var}Y = \frac{1}{(a-1)^2(a-2)b^2}$.

d. $f_x(x) = \frac{1}{\Gamma(3/2)\beta^{3/2}}x^{3/2-1}e^{-x/\beta}$, $x > 0$. For $Y = (X/\beta)^{1/2}$, $f_Y(y) = \frac{2}{\Gamma(3/2)}y^2e^{-y^2}$, $y > 0$. To calculate the moments we use integration-by-parts with $u = y^2$, $dv = ye^{-y^2}$ to obtain

$$\mathrm{E}Y = \frac{2}{\Gamma(3/2)}\int_0^\infty y^3e^{-y^2}dy = \frac{2}{\Gamma(3/2)}\int_0^\infty ye^{-y^2}dy = \frac{1}{\Gamma(3/2)}$$

and with $u = y^3, dv = ye^{-y^2}$ to obtain

$$\mathrm{E}Y^2 = \frac{2}{\Gamma(3/2)}\int_0^\infty y^4e^{-y^2}dy = \frac{3}{\Gamma(3/2)}\int_0^\infty y^2e^{-y^2}dy = \frac{3}{\Gamma(3/2)}\sqrt{\pi}.$$

Using the fact that $\frac{1}{2\sqrt{\pi}}\int_{-\infty}^\infty y^2e^{-y^2} = 1$, since it is the variance of a n$(0, 2)$, symmetry yields $\int_0^\infty y^2e^{-y^2}dy = \sqrt{\pi}$. Thus, $\mathrm{Var}Y = 6 - 4/\pi$, using $\Gamma(3/2) = \frac{1}{2}\sqrt{\pi}$.

e. $f_x(x) = e^{-x}$, $x > 0$. For $Y = \alpha - \gamma\log X$, $f_Y(y) = e^{-e^{\frac{\alpha-y}{\gamma}}}e^{\frac{\alpha-y}{\gamma}}\frac{1}{\gamma}$, $-\infty < y < \infty$. Calculation of $\mathrm{E}Y$ and $\mathrm{E}Y^2$ cannot be done in closed form. If we define

$$I_1 = \int_0^\infty \log x e^{-x}dx, \qquad I_2 = \int_0^\infty (\log x)^2 e^{-x}dx,$$

then $\mathrm{E}Y = \mathrm{E}(\alpha - \gamma\log x) = \alpha - \gamma I_1$, and $\mathrm{E}Y^2 = \mathrm{E}(\alpha - \gamma\log x)^2 = \alpha^2 - 2\alpha\gamma I_1 + \gamma^2 I_2$.The constant $I_1 = .5772157$ is called Euler's constant.

16.

Let $Y=$ # of R.V. that exceed $1/2$.

Then $Y \sim \text{Binomial}(3, P(X_i \geq 1/2))$

$P(X > 1/2) = \int_{1/2}^{1} 3x^2\,dx = x^3\Big|_{1/2}^{1} = 1-(1/2)^3 = 7/8$

$P(Y=2) = \binom{3}{2}\left(\frac{7}{8}\right)^2\left(\frac{1}{8}\right)^1 = 0.287$

17. (a) $f_X(x) = \int_0^\infty e^{-(x+y)}\,dy = -e^{-(x+y)}\Big|_0^\infty = \boxed{e^{-x} = f_X(x) \quad x>0}$

$\boxed{f_Y(y) = e^{-y} \quad y>0}$

(b) $\int_0^y\int_0^x e^{-(s+t)}\,ds\,dt = \int_0^y \left[-e^{-(s+t)}\Big|_0^x\right]dt = \int_0^y \left[e^{-t} - e^{-(x+t)}\right]dt = -e^{-t} + e^{-(x+t)}\Big|_0^y =$

$= -e^{-y} + e^{-(x+y)} - (-1+e^{-x}) = 1-e^{-x}-e^{-y}+e^{-(x+y)} = \boxed{(1-e^{-x})(1-e^{-y}) \quad x>0,\ y>0}$

(c) $P(X>2) = \int_2^\infty e^{-x}\,dx = -e^{-x}\Big|_2^\infty = \boxed{e^{-2}}$

(d) $P(X<Y) = \int_0^\infty\int_0^y e^{-(x+y)}\,dx\,dy = \int_0^\infty \left[-e^{-(x+y)}\Big|_0^y\right]dy = \int_0^\infty e^{-y} - e^{-2y}\,dy = -e^{-y} + \frac{1}{2}e^{-2y}\Big|_0^\infty = 1-\frac{1}{2} = \boxed{\frac{1}{2}}$

(e) $P(X+Y>2) = P(X>2-Y) = 1-P(X\leq 2-Y) = 1-\int_0^2\int_0^{2-y} e^{-(x+y)}\,dx\,dy = 1-\int_0^2\left[-e^{-(x+y)}\Big|_0^{2-y}\right]dy$

$= 1-\int_0^2\left[-e^{-(2-y+y)} + e^{-y}\right]dy = 1-\int_0^2 e^{-y} - e^{-2}\,dy = 1-\left[-e^{-y} - ye^{-2}\Big|_0^2\right] = 1-\left[-e^{-2}-2e^{-2}+1\right]$

$= \boxed{3e^{-2}}$

(f) Yes, because $f_{X,Y}(x,y) = e^{-(x+y)} = e^{-x}e^{-y} = f_X(x)f_Y(y)$